Wk12L2: Magnetically Coupled Circuits

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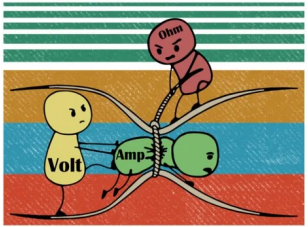

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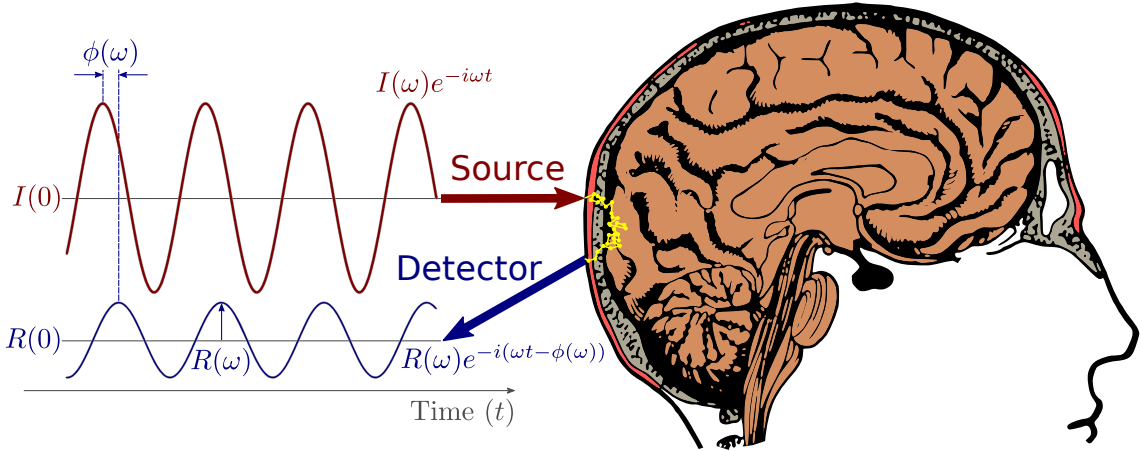
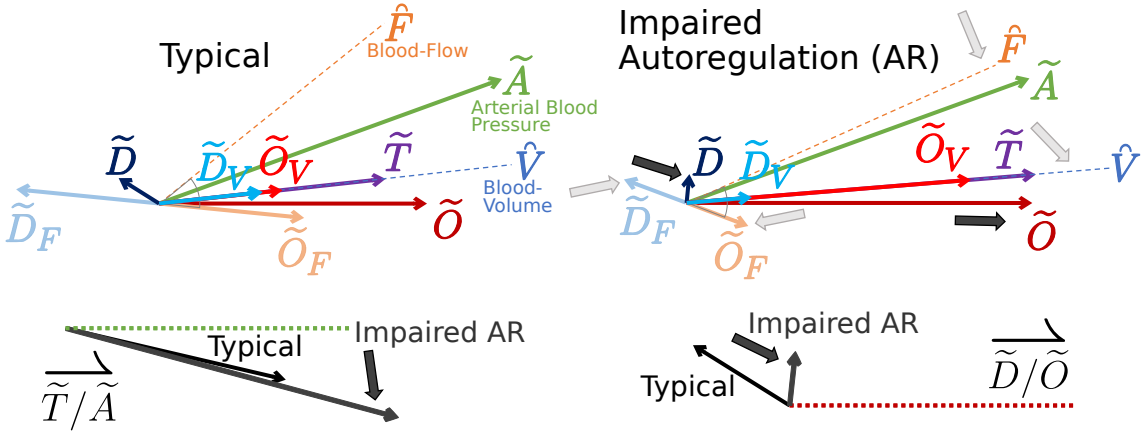
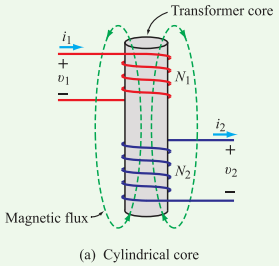
Some Transfer Function Applications 1/2

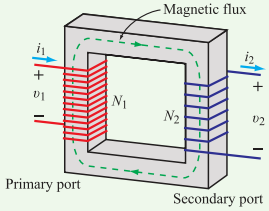


Some Transfer Function Applications 2/2





(a) Cylindrical core



(b) Rectangular core

Figure 11-7: Magnetically coupled coils.

- **Turns ratio** (n):

$$n = N_2/N_1$$

$$L \propto N^2 \Rightarrow n^2 = L_2/L_1$$

- Voltage ratio:

$$\widetilde{V}_2/\widetilde{V}_1 = n$$

- Current ratio:

$$\widetilde{I}_2/\widetilde{I}_1 = 1/n$$

- Transformer types:
  - **Step-up**: $n > 1$ & $|\widetilde{V}_2| > |\widetilde{V}_1|$
  - **Step-down**: $n < 1$ & $|\widetilde{V}_2| < |\widetilde{V}_1|$

Reminders

- Recitation Friday